HANDIKO GESANG ANUGRAH SEJATI

IoT Firmware & Hardware Engineer
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Portfolio: handiko.github.io | GitHub: github.com/handiko

PROFESSIONAL SUMMARY

Innovative Firmware Engineer with over 5 years of experience in the research and development of IoT systems, RF communication protocols, and industrial hardware. Proven expertise in designing complex firmware for maritime AIS systems and automated industrial feeders. Specialized in bridging high-level algorithms (Python/C++) with low-level hardware constraints (ESP12, STM32, Arduino). Strong background in RF design, PCB development, and MQTT-based IoT architectures.

CORE COMPETENCIES

- **Embedded Systems:** Firmware Development (C/C++), STM32MCU, STM-HAL, Low Power Optimization.
- **IoT & Connectivity:** BLE (STM32W, ESP32, HM-10/JDY-08), Wi-Fi (ESP8266/ESP32), GPS/NMEA.
- **RF & Communication:** AIS (Automatic Identification System), APRS, Software Defined Radio (SDR), Signal Processing, Antenna Configuration.
- **Hardware Design:** PCB Layout (EasyEDA), Circuit Design, Sensor Integration, RF (Radio Frequency) Circuit Design.
- **Languages & Tools:** Embedded C/C++, Python, Git.

PROFESSIONAL EXPERIENCE

PT. Imani Prima | *Firmware Engineer* | **2020 – 2025**

- **Maritime IoT:** Spearhead R&D for AIS (Automatic Identification System) transmitters and receivers used by national maritime and fisheries institutions.
- **Firmware Architecture:** Developed robust firmware to track mobile assets (vessels/barges) under challenging maritime environments.
- **RF Integration:** Optimized transceiver hardware design to improve signal reliability and maritime traffic monitoring.

PT. Multidaya Teknologi Nusantara (eFishery) | *Electrical Hardware Engineer* | **2019 – 2020**

- **Industrial Automation:** Designed hardware for automated e-feeder units, utilizing physics-based modeling to predict fish food distribution.

- **Algorithm Development:** Created a proprietary motor-current measurement algorithm to accurately detect and report food dispense volume via IoT.
- **Sensor Fusion:** Integrated environmental sensors into a centralized IoT ecosystem for real-time aquaculture monitoring.

PT. Integrasi Sinergi Teknologi (INSITEK) | *Intern* | **2018**

- **Wireless Infrastructure:** Researched antenna height and gain configurations for hot-air-balloon-based internet hotspots.

KEY PROJECTS

IoT-Enabled AIS Receiver & Gateway

IoT-based Automatic Identification System (AIS) Receiver designed to capture marine vessel data and relay it to centralized monitoring servers. Built around an ESP32 and an STM32F4, the system features a web-based configuration interface for easy deployment and management.

Low-Cost AIS Class-B Transmitter for Small Vessels

A budget-friendly Class-B AIS (Automatic Identification System) Transmitter designed to provide situational awareness for small boats. By utilizing common, off-the-shelf microcontrollers and RF components, this solution ensures supply chain resilience and affordability for maritime operators who find commercial AIS units cost-prohibitive.

EDUCATION

Universitas Gadjah Mada (UGM) | *Bachelor of Engineering in Engineering Physics*

- **Specialization:** Instrumentation and Control.
- **Final Project:** Multi-channel modulators for Software Defined Radio.

TECHNICAL SKILLS

- **Microcontrollers:** ESP32, STM32, ATmega (Arduino).
- **Protocols:** MQTT, I2C, SPI, UART, BLE, RF protocols (AIS, APRS).
- **Design Tools:** EasyEDA, STMCubeIDE, Arduino, GNU Radio, VS Code, PlatformIO.
- **Programming:** C/C++ (Embedded), Python (Data & Tools).